

Chapter 4

Inductors and Capacitors

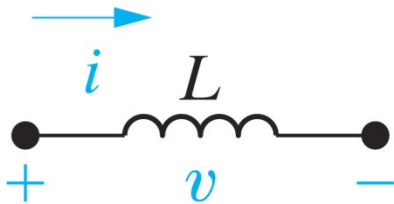
Inductors

- Based on Magnetic Field Phenomena
 - Stores energy *
- Passive Element

- *Moving Charge $\equiv$ Current
- If i varies with time, the magnetic field varies with time

For circuit analysis, we need the Current – Voltage Relationship

Schematic Symbol: Coil



$L \equiv$ Inductance
Units $\equiv$ Henrys (H)

$$v = L \frac{di}{dt}$$

Compare to Ohm's Law

$$v = Ri$$

Notes on Inductors

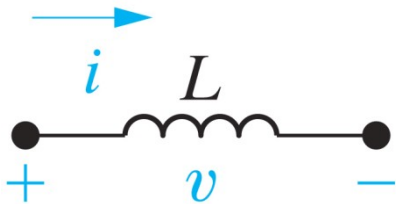
v is proportional to time rate-of-change of i

1. For Constant (DC) Current, $v = 0$
 $\therefore$ **Inductor is a “Short” for DC**

$$v = L \frac{di}{dt}$$

1. Current Can't Change Instantly;
Would require ∞ Voltage

$$v = L \frac{di}{dt}$$



$$L = \frac{\mu N^2 A}{S}$$

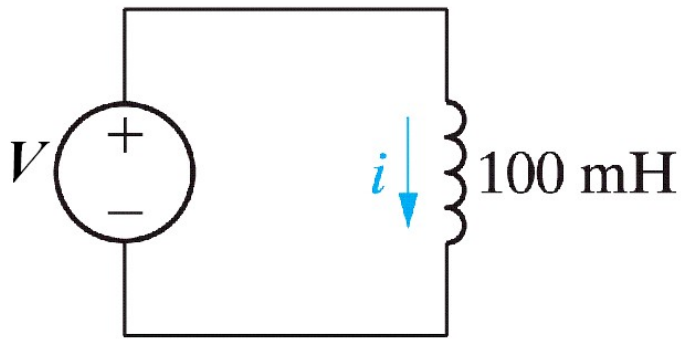
μ = Permeability

N = Number of Turns

A = Cross-Sectional Area

S = Axial Length

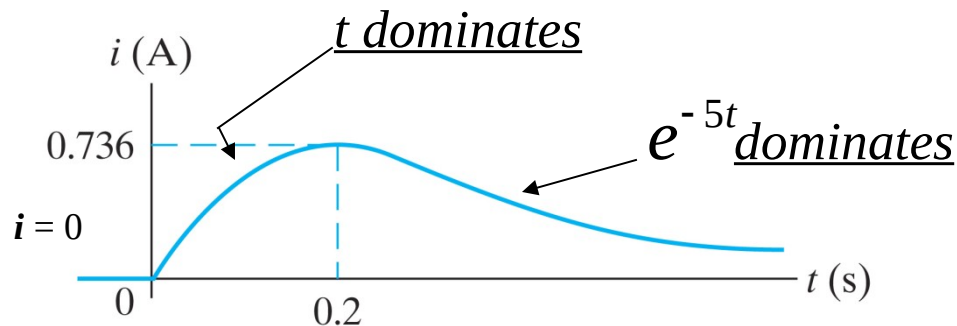
Current in an Inductor



$$\left\{ \begin{array}{l} V = e^{-5t} (1 - 5t) \quad t \geq 0 \\ i = 10te^{-5t} \quad t \geq 0 \end{array} \right\} \quad \text{Given}$$

Must Confirm to $v = L \frac{di}{dt}$

Sketch $i(t)$



Sketch $v(t)$

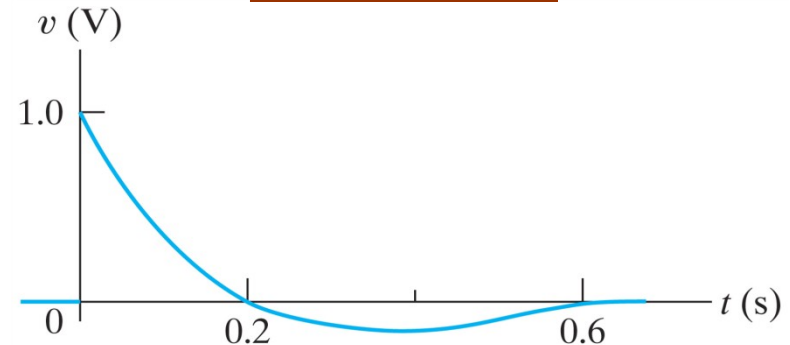


Figure: 06-04Ex6.1

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Initially, t dominates, i increases.

Eventually, $1/e^{5t}$ dominates, i decreases

Current in an Inductor (Contd.)

**Find
Maximum
Current in
an Inductor**

$$\frac{di}{dt} = 10 \left[e^{-5t} + t(-5e^{-5t}) \right] \equiv 0$$

**Take derivative and set
to zero**

$$= 10e^{-5t} [1 - 5t] = 0$$

Simplify

$$e^{-5t} [1 - 5t] = 0$$

$$\therefore \frac{di}{dt} = 0 \quad \text{when}$$

$$t = \frac{1}{5} = 0.2 \text{ (s)}$$

$$i(t = 0.2) = 0.736 \text{ (A)}$$

Maximum value

Current in an Inductor (Contd.)

Derive $v(t)$ from $i(t)$

$$L = 100\text{mH} \quad \left\{ v = L \frac{di}{dt} = 0.1 [10e^{-5t} (1 - 5t)] \right\}$$

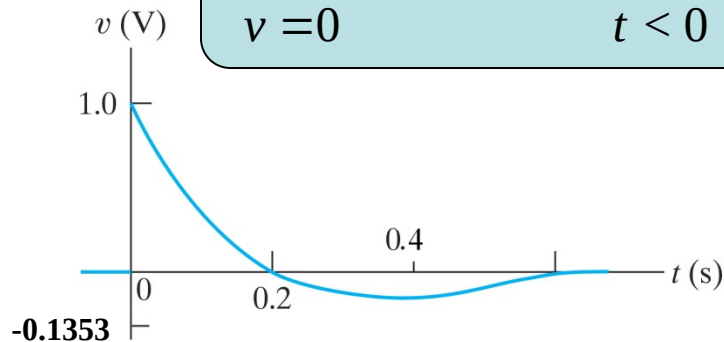
$$i(t) = 0 \text{ for } t < 0 \quad i(t) = 10te^{-5t} \text{ for } t \geq 0$$

Take derivative of $i(t)$

$i(t)$ is Continuous

$$\begin{aligned} v &= e^{-5t} (1 - 5t) & t \geq 0 \\ v &= 0 & t < 0 \end{aligned}$$

Voltage across inductor



Find Minimum

$$\frac{\partial v}{\partial t} = -5e^{-5t} (1 - 5t) + e^{-5t} (-5)$$

$$= 5e^{-5t} (5t - 2) = 0$$

$$t = 0.4\text{s} \quad v_{\min}(0.4) = -0.1353\text{(V)}$$

$$\left\{ \begin{aligned} v(0) &= e^0(1-0) = 1\text{(V)} \\ v &= 0 & \text{when } t = 0.2\text{s} \\ v &= 0 & \text{when } t \rightarrow \infty \end{aligned} \right\}$$

$v(t)$ is Discontinuous

Take derivative and set to zero

Analytic Expression for Current in an Inductor

$$\left. v = L \frac{di}{dt} \right\} \text{Integrate equation}$$

$$L di = v dt$$

L is Constant $L \int di = \int v d\tau + C$ $\longleftarrow$ Integration Constant

$$i(t) = \frac{1}{L} \int v d\tau + C' \quad \left[C' = \frac{C}{L} \right]$$

Lets say τ ranges from $t_0 \rightarrow t$

$$\therefore i(t) = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + C'$$

$$@ \ t = t_0 \text{ (initial point in time), } i(t) \equiv i(t_0), \Rightarrow \int_{t_0}^{t_0} v(\tau) d\tau = 0$$

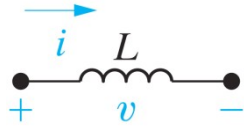
$$\therefore C' = i(t_0)$$

$$\therefore i(t) = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0)$$

Notes on Inductor Equations

- $$i(t) = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0)$$
 - * $i(t_0)$ is initial current
 - * Often $t_0 \equiv 0 \Rightarrow i(0)$
 - * Often $i(0)$ or $i(t_0) = 0$
No initial current
- $$v = L \frac{di}{dt}$$
 - * Current Can't Change Instantly
 - * Current Can be Stored

Power and Energy



$$p = Vi = Li \frac{di}{dt}$$

Power

$$w = \int p dt$$

Energy definition

$$w = \int Li \frac{di}{dt} dt = \int L i di$$

Substitute Power Expression

$$w = L \left[\frac{1}{2} i^2 \right]$$

Integrate

$$\therefore w = \frac{1}{2} Li^2$$

Energy in Inductor

Derivative of w
w.r.t. time

$$\left\{ \frac{dw}{dt} = \frac{1}{2} L(2i) \frac{di}{dt} = Li \frac{di}{dt} = p \right.$$

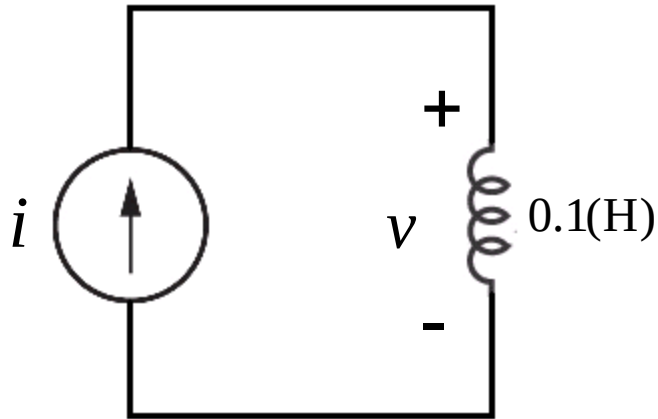
Power

$$p = Vi$$

$$V = L \frac{di}{dt}$$

$$i = \frac{1}{L} \int_0^t v d\tau + i(t_0)$$

Power and Work Graphs



$$\left. \begin{array}{ll} i = 0, & t < 0 \\ i = 10t \exp(-5t) & t \geq 0 \end{array} \right\}$$

$$\left\{ \begin{array}{l} v = L \frac{di}{dt} \\ p = vi \\ w = \frac{1}{2} Li^2 \end{array} \right\}$$

Curves come from these equations.

Notes on Curves

- ① Peak in $i(t)$ corresponds to $v(t) = 0$, since $i_{\max}$ occurs when $di/dt = 0$
- ② $v(t) \equiv \text{Negative}$ when $di/dt \equiv \text{Slope} \equiv \text{Negative}$
- ③ **Power = Voltage $\times$ Current (Competing Processes)**
 $\left\{ \begin{array}{cc} \downarrow & \uparrow \end{array} \right\}$

Power and Work Graphs (Contd.)

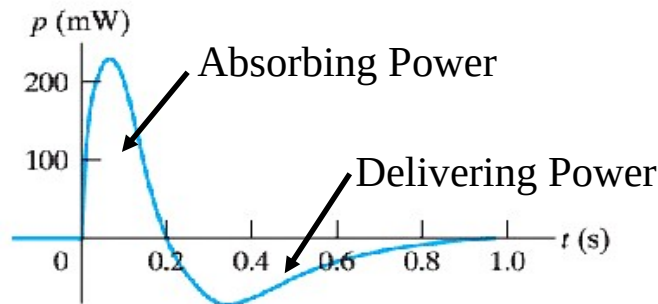
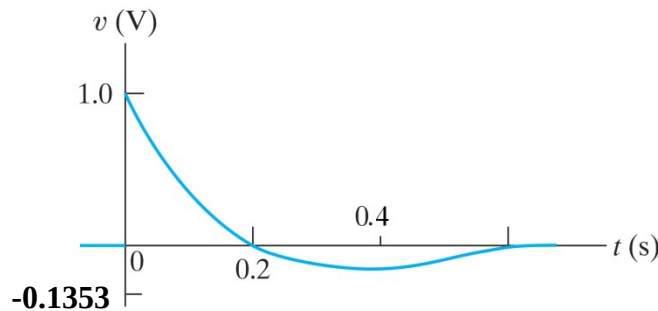
$$p = 0 \text{ when } v = 0$$

$$p < 0 \text{ (negative) when } v < 0$$

Power peaks when decrease in v overwhelms increase in i

④ Power and Energy Comparison

Time	Power	Energy
$t = 0$	$p = 0$	$W = 0$
$0 \leq t \leq 0.2$	$p > 0$ “Absorbed”	Increasing; “Stored”
$0.2 \leq t \leq \infty$	$p < 0$ “Delivered”	Decreasing; “Extracted”



$$p = vi$$

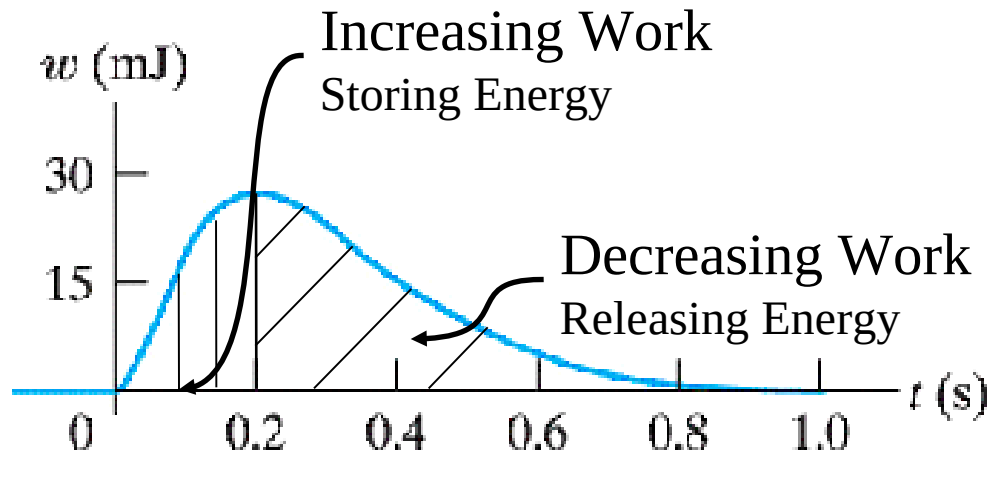
$$i(t) \geq 0 \text{ for } t > 0$$

when $v > 0$ absorbing power

when $v < 0$ delivering power

Power and Work Graphs (Contd.)

$$W = \int p dt$$

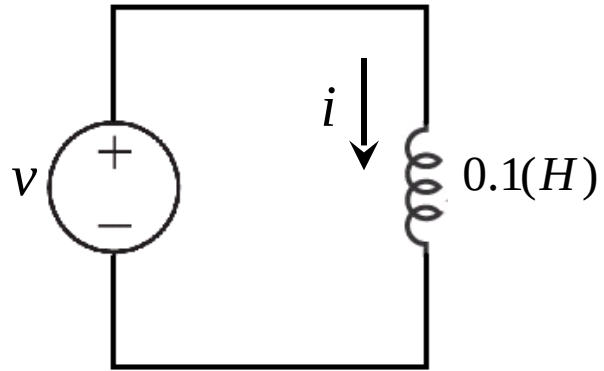


Integrate
Power from
the previous
slide

$$\textcircled{5} \quad W = \frac{1}{2} Li^2$$

w is Maximum when i is Maximum

Example: Graph the Power and Work



$$v = 0 \quad \text{for } t < 0$$

$$v = 20t e^{-10t} \quad \text{for } t \geq 0$$

$$\left\{ \begin{array}{l} i = \frac{1}{L} \int_0^t v d\tau + i(t_0) \\ p = vi \\ w = \frac{1}{2} Li^2 \end{array} \right\} \quad \begin{array}{l} i(t_0) = 0 \\ \text{Given} \end{array}$$

$$i = \frac{1}{0.1} \int_0^t 20\tau e^{-10\tau} d\tau + 0$$

$$\left\{ \begin{array}{l} \text{Integrate by parts} \\ \text{or} \\ \text{Use integral tables} \end{array} \right\}$$

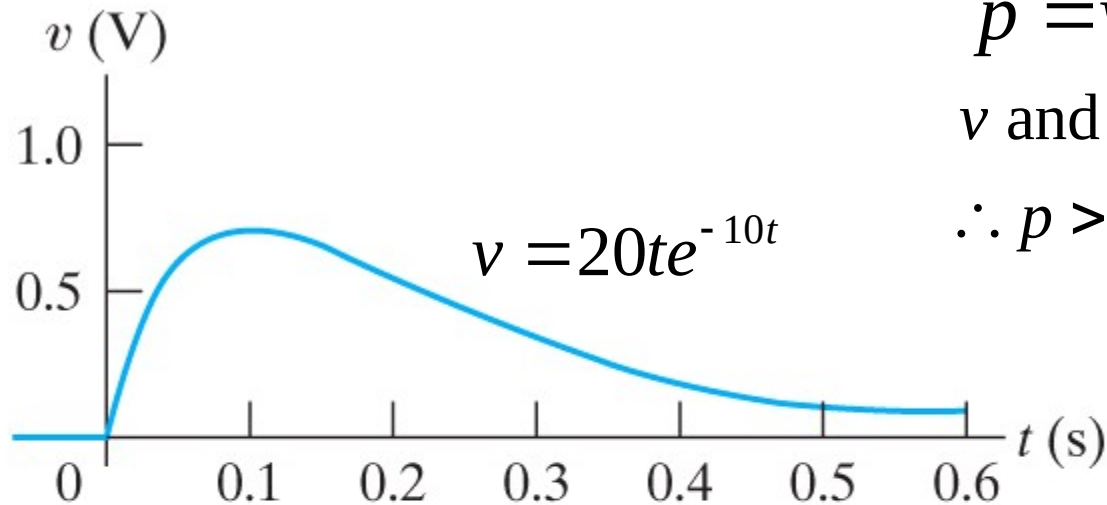
$$i = 2(1 - 10te^{-10t} - e^{-10t}) \quad t \geq 0$$

Analytical expression for $i(t)$

$$i(t=0) = 0$$

$$i(t \rightarrow \infty) \rightarrow 2(A)$$

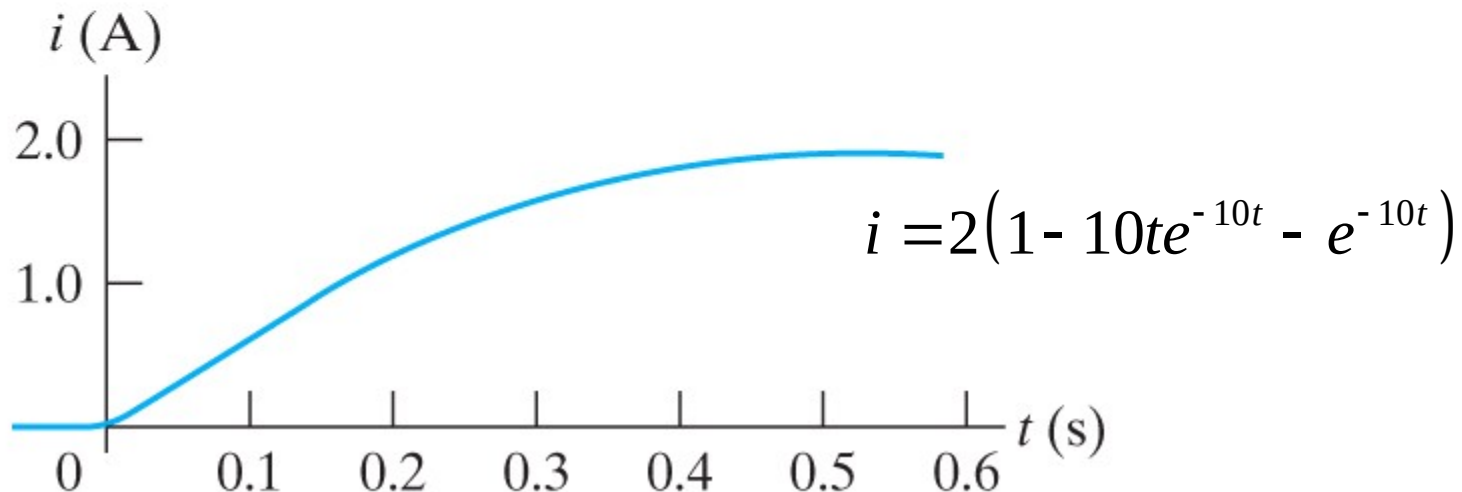
Graph of Voltage and Current



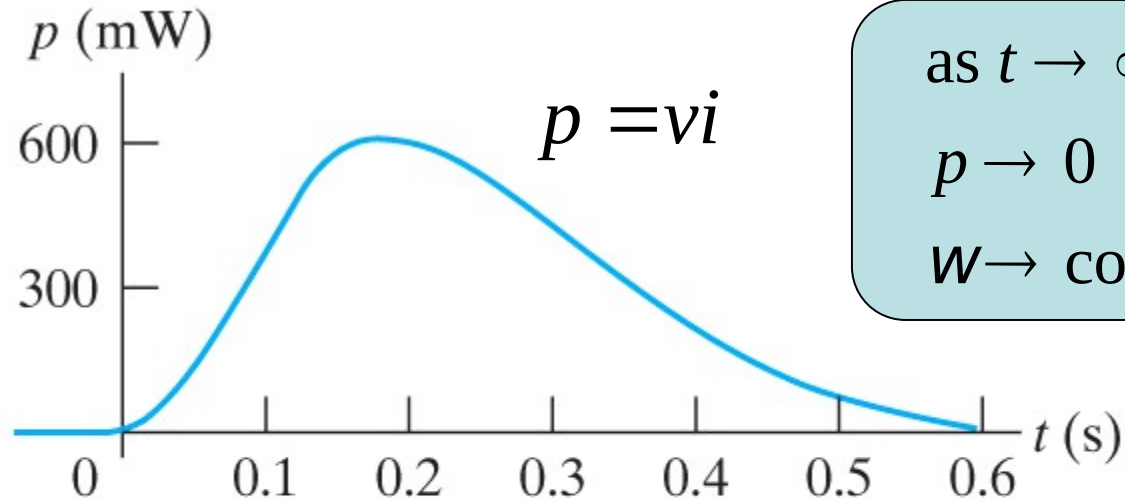
$$p = vi$$

v and i are always positive

$\therefore p > 0 \Rightarrow$ Power is absorbed



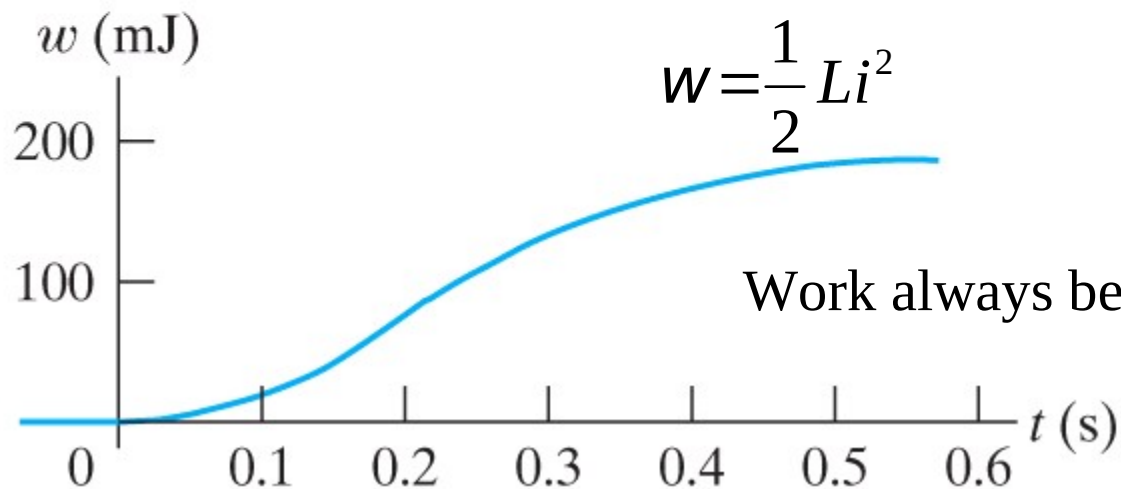
Graph of Power and Work



as $t \rightarrow \infty$

$p \rightarrow 0$

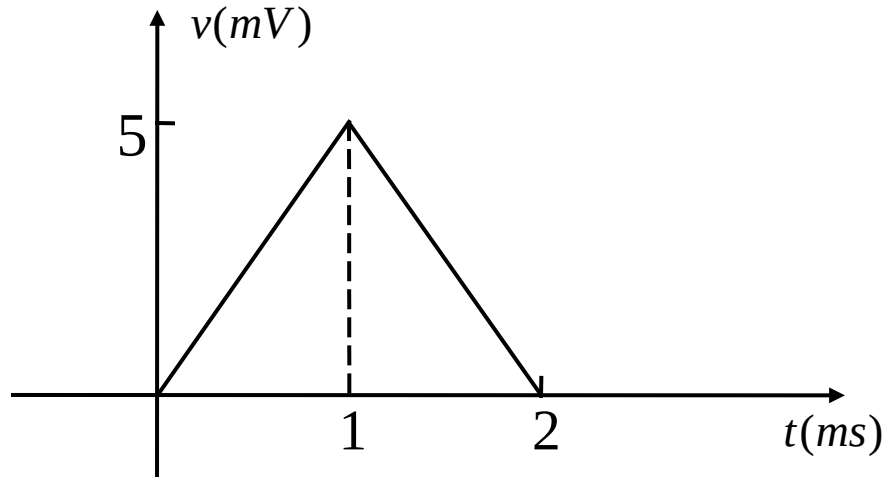
$W \rightarrow \text{constant}$



$$w = \frac{1}{2} Li^2$$

Work always being stored

Math Review



How do you write an expression for $v(t)$?

$$\left. \begin{array}{ll} v(t) = 0 & t < 0 \\ v(t) = 0 & t > 2(ms) \end{array} \right\} \text{Obvious}$$

- $(0 \leq t \leq 1(ms))$ $y = mx + b$
 - slope m
 - y-intercept b
$$m = \frac{\text{Rise}}{\text{Run}} = \frac{5mV}{1mV} = 5$$

$$b = 0$$

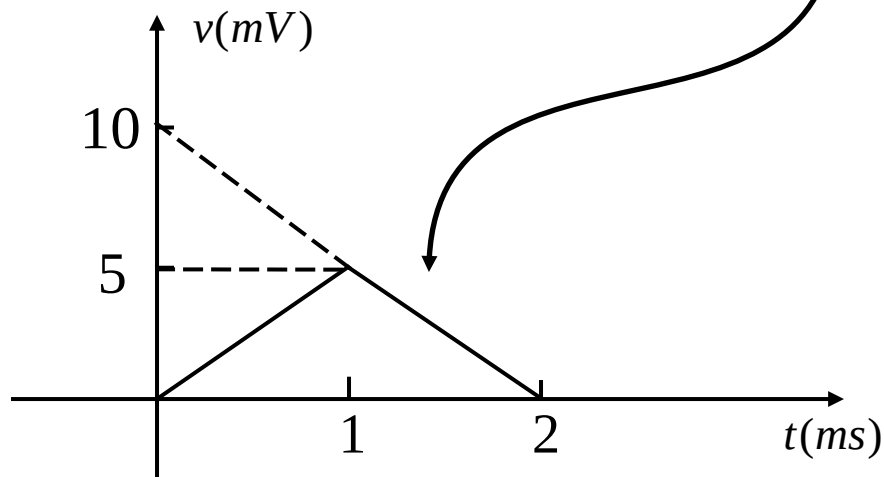
$$v(t) = 5t \quad (0 \leq t \leq 1(ms))$$

Math Review (Contd.)

- $(1(ms) \leq t \leq 2(ms))$

$$m = - \frac{5(mV)}{(2 - 1)(ms)} = -5$$

From the figure



$$b = 10(mV) = 0.01(V)$$

$$\therefore v(t) = -5t + 0.01 \text{ (Volts)}$$

From the above figure
for $1(ms) \leq t \leq 2(ms)$

Math Review (Contd.)

Sine Waves

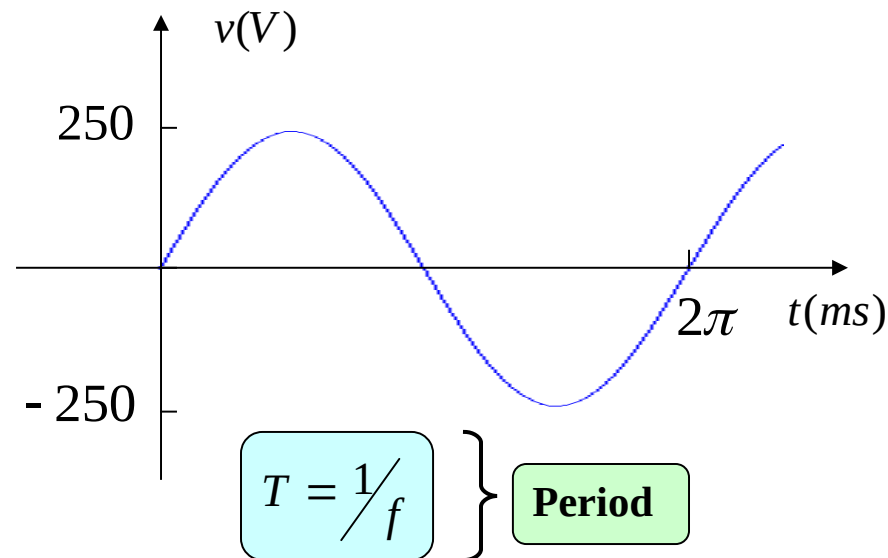
$$v = 250 \sin(1000t)$$

$$v = V_m \sin(\omega t)$$

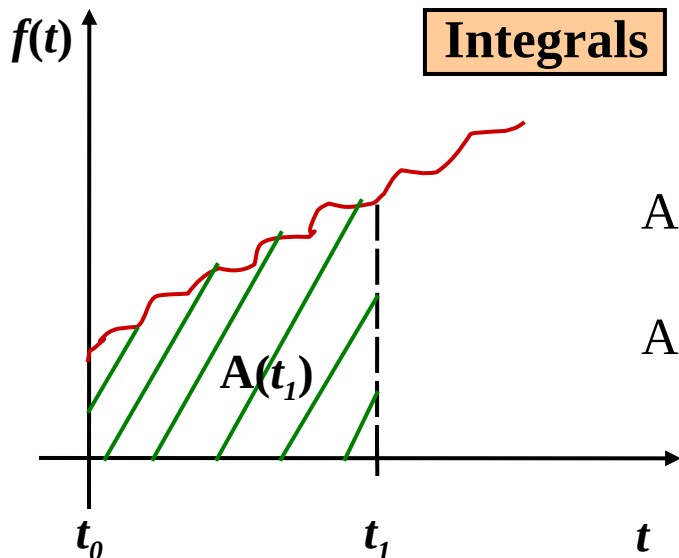
$$V_m = 250$$

$$\omega = 1000 = 2\pi f$$

$$\text{Hz} \left\{ f = \frac{1000}{2\pi} \quad T = \frac{2\pi}{1000} = 2\pi(\text{ms}) \right.$$



Integrals



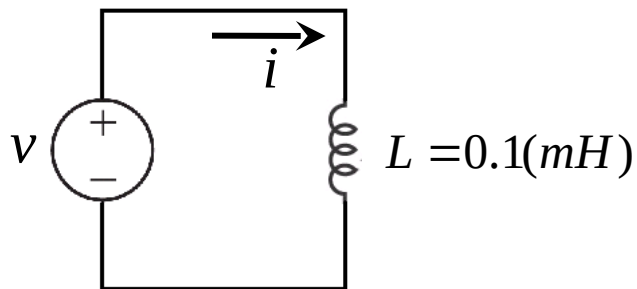
$$A(t_1) = \int_0^{t_1} f(\tau) d\tau = \text{Specific Area}$$

$$A(t) = \int_0^t f(\tau) d\tau = A \equiv \text{Function of } t$$

Upper limit is a constant

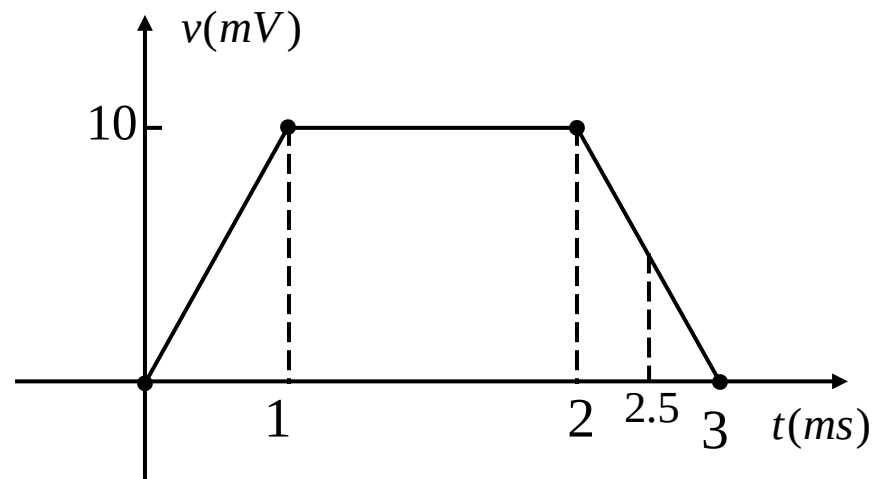
Upper limit is a variable

Inductor Example: Given $v(t)$, Find $i(2.5\text{ ms})$



Given $i(0) = 0$ $\frac{1}{L} = 10,000$

Assume $v(t) = 0$ $t \leq 0$



$$y = mx + b$$

$$\left[\begin{array}{l} m = \frac{10(\text{mV})}{1(\text{ms})} = 10 \\ b = 0 \end{array} \right]$$

$$v(t) = 10t(\text{V}) \quad (0(\text{ms}) \leq t \leq 1(\text{ms}))$$

$$v(t) = 10\text{mV} = 0.01(\text{V}) \quad (1(\text{ms}) \leq t \leq 2(\text{ms}))$$

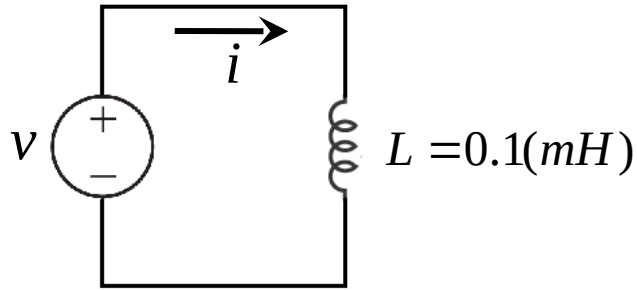
$$v(t) = -10t + 0.03(\text{V}) \quad (2(\text{ms}) \leq t \leq 3(\text{ms}))$$

$$v(t) = 0 \quad (t \geq 3(\text{ms}))$$

$$\left[\begin{array}{l} m = -\frac{10(\text{mV})}{(3-2)(\text{ms})} = -10 \\ b = 10 + 10 + 10 = 30(\text{mV}) \end{array} \right]$$

**Mathematical
Expression
for
 $v(t)$**

Inductor Example: Find $i(1ms)$ (Contd.)



$$i(t) = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0)$$

Formula for
current in an
inductor

$$\textcircled{1} (0(ms) \leq t \leq 1(ms)) : i(t_0) = 0 \quad t_0 = 0$$

$$i(1ms) = \frac{1}{L} \int_0^{1(ms)} v(\tau) d\tau = \frac{1}{L} \text{Area} \Big|_0^{1(ms)} = 10,000 \left[\frac{1}{2} (\text{base})(\text{height}) \right]$$

Formula
for
triangle
area

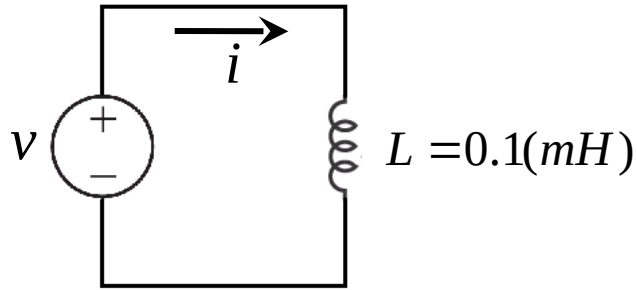
$$i(1ms) = 10,000 \left(\frac{1}{2} [0.001][0.01] = 0.05 \right)$$

Simplify

$$\therefore i(1ms) = 50(mA)$$

We “Charged” L for $1(ms)$, and now it has a
“stored” current of $50(mA)$

Inductor Example: Find $i(2ms)$ (Contd.)



$$\textcircled{2} \quad (1(ms) \leq t \leq 2(ms)) \quad t_0 = 1(ms) \quad i(t_0) = i(1ms) = 50(mA)$$

From the previous slide

$$i(2ms) = \frac{1}{L} \int_{1(ms)}^{2(ms)} v(\tau) d\tau + i(1) \quad \left\{ \text{Use formula again} \right\}$$

$$i(2ms) = \frac{1}{L} \text{Area} \Big|_{1(ms)}^{2(ms)} + 50mA = 10,000 [\text{base} \times \text{height}] + 50(mA) \quad \left\{ \text{Formula for rectangle area} \right\}$$

$$i(2ms) = 10,000[(0.001)(0.01)] + 0.05$$

$$i(2ms) = (0.1 + 0.05)(A) \quad \left\{ \text{Plug in the numbers} \right\}$$

$$i(2ms) = 0.15(A) = 150(mA)$$

After 2(ms), we have 150(mA) stored

Inductor Example: Find $i(2.5ms)$ and $i(3ms)$ (Contd.)

$$\textcircled{3} \quad (2(ms) \leq t \leq 3(ms)) \quad t_0 = 2(ms) \quad i(t_0) = i(2ms) = 150(mA)$$

$$i(t) = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0) \quad \left\{ \text{General formula} \right.$$

$$\int_{2ms}^t (-10\tau + 0.03) d\tau = \left[-\frac{10\tau^2}{2} + 0.03\tau \right] \bigg|_{2ms}^t \quad \left\{ \text{Plug in analytical expression for } v(t) \right.$$

$$= \left[(-5t^2 + 0.03t) - (-5[0.002]^2 + 0.03[0.002]) \right] \quad \left\{ \text{Plug in limits} \right.$$

$$= -5t^2 + 0.03t - [4 \times 10^{-5}] \quad \left\{ \text{Simplify} \right.$$

$$i(t) = 10,000[-5t^2 + 0.03t - 4 \times 10^{-5}] + \underbrace{0.15}_{i(t_0)} \quad \left\{ \text{Add initial conditions} \right.$$

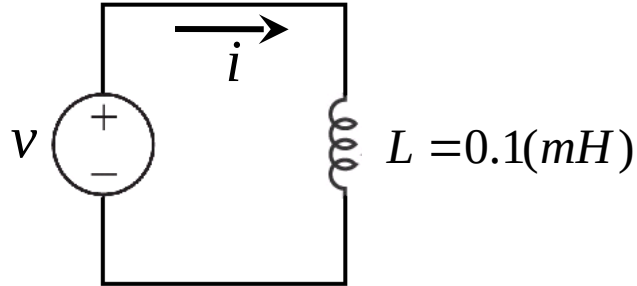
$$i(t) = -50,000t^2 + 300t - 0.4 + 0.15 \quad \left\{ \text{Simplify} \right.$$

$$i(t) = -50,000t^2 + 300t - 0.25 \quad \left\{ \begin{array}{l} \text{General expression for } i(t) \\ \text{for } 2(ms) \leq t \leq 3(ms) \end{array} \right.$$

$$i(2.5ms) = 0.1875(A) = 187.5(mA) \quad \left\{ \text{Let } t = 2.5(ms) \right.$$

$$i(3ms) = 0.2(A) = 200(mA) \quad \left\{ \text{Let } t = 3(ms) \right.$$

Inductor Example (Contd.)



④ What about $t > 3(ms)$?

$$(3(ms) \leq t \leq \infty) \quad i(t) = \frac{1}{L} \int_0^t v(\tau) d\tau + i(t_0) \quad \left. \vphantom{\int_0^t} \right\} \text{General formula}$$

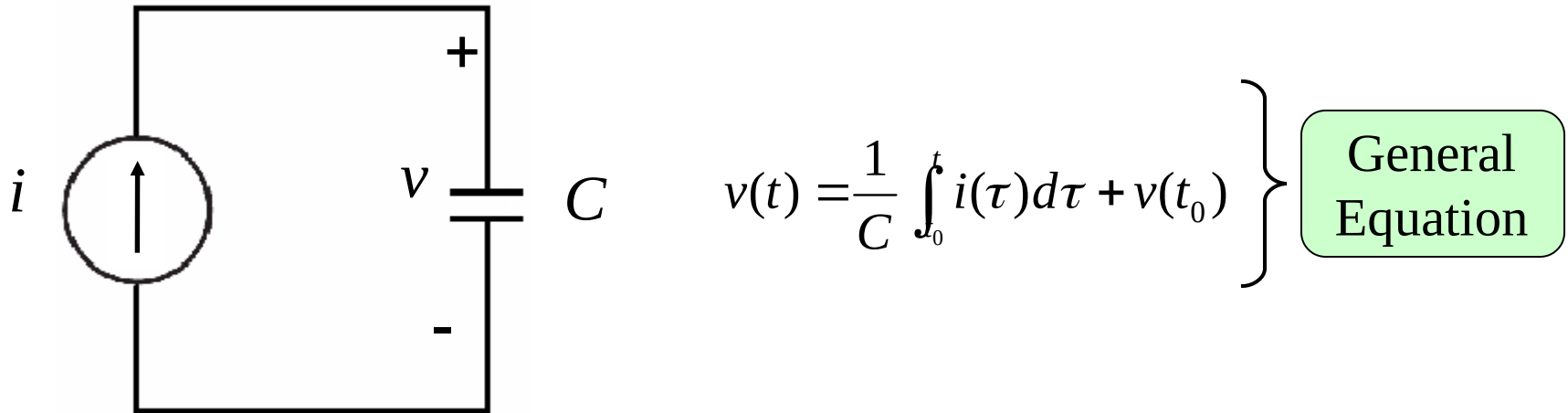
$$v(t) = 0 \quad \therefore \int v(\tau) d\tau = 0$$

$$t_0 = 3(ms)$$

$$\therefore i(t) = i(3ms) = 200(mA) \quad \text{for } 3(ms) \leq t \leq \infty$$

- Current in the Inductor Remains Constant at 200(mA) from $t = 3(ms)$ $\longrightarrow$ Forever
- Stored Current gives Stored Energy

Dual Problem for Capacitors



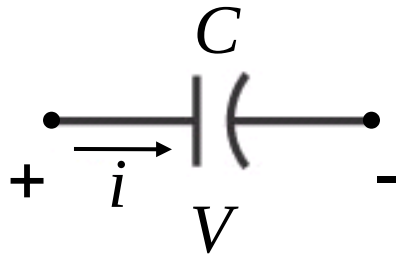
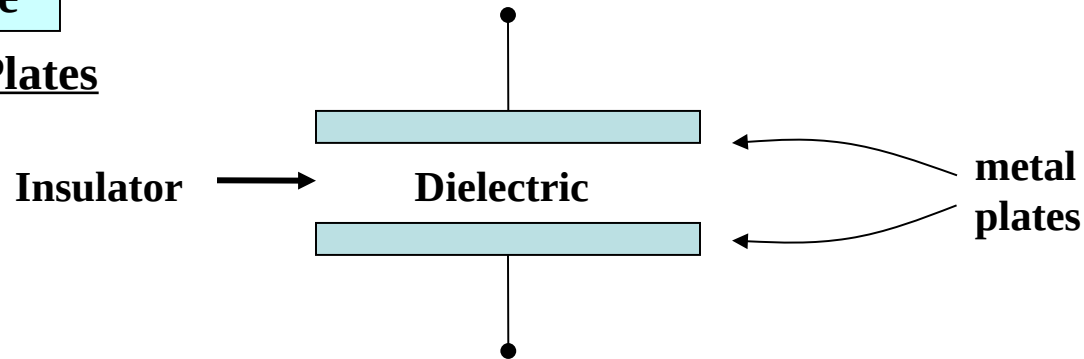
- Use Same Procedure as for the Inductor Example
- Capacitor Charges Up To a Constant Voltage
- Stored Charge $\equiv$ Memory

Capacitor

Electric Field Phenomena

Stores Charge

Parallel Plates



$C \equiv$ Capacitance

Units = Farad (F)

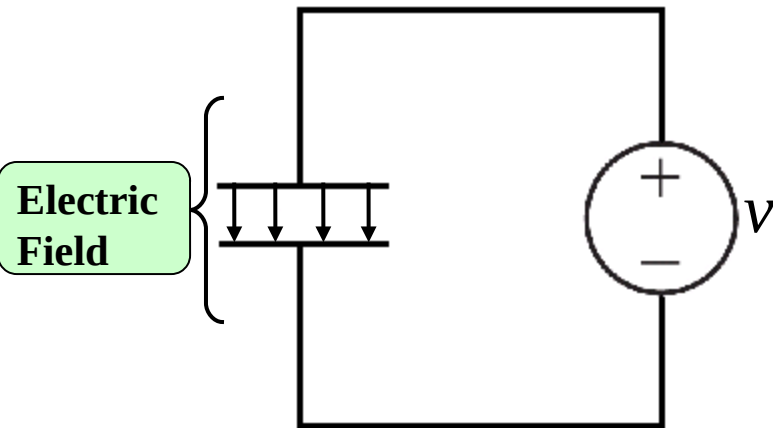
Normally $\mu F, pF$

$$\left\{ C = \frac{\epsilon_0 \epsilon_r A}{D} \right\}$$

$A = \text{Area}$
 $D = \text{Thickness}$
 $\epsilon_0 = \text{Permittivity of free space}$
 $\epsilon_r = \text{Relative Permittivity or Dielectric Constant}$

Insulator “Ideally” Prevents Charge from Flowing Through the Device

Where Does Current Come From in a Capacitor?



- Charge Stored On Each Plate Equal And Opposite
- V Changes, Q Changes
- $\Delta Q \rightarrow$ Displacement Current

- Displacement Current is Indistinguishable From Conduction Current.
- If Voltage is Constant, Charge is Constant, $i = 0$.

$$i = C \frac{dv}{dt}$$

$$\left\{ \begin{array}{l} \text{Compare to Ohm's Law} \\ i = \left(\frac{1}{R} \right) v \end{array} \right\}$$

Capacitors Notes:

i is related to a time rate-of-change in v

① For Constant (DC) Voltage $\Rightarrow i = 0$ } $i = C \frac{dv}{dt}$

< Only Time Varying Voltage $\rightarrow$ Displacement Current. >

$\therefore$ Capacitor is an “Open” for DC

② Voltage Can't Change Instantly } $i = C \frac{dv}{dt}$

* Would Require ∞ Current *

Voltage Across Capacitor: < Similar to Procedure for Inductor >

$v(t) = \frac{1}{C} \int_0^t i(\tau) d\tau + v(t_0)$ } $\text{Integrate } i = C \frac{dv}{dt}$

Power and Energy for a Capacitor

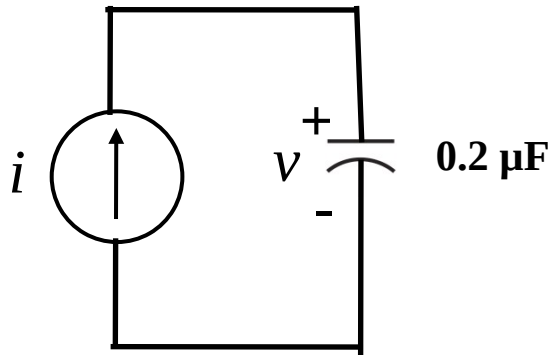
$$\left\{ \begin{array}{l} \text{Power} \\ p = vi = Cv \frac{dv}{dt} \end{array} \right. \quad \left\{ \begin{array}{l} W = \frac{1}{2} C v^2 \\ \text{Energy} \end{array} \right.$$

Duality Between L and C

Inductor	Capacitor
Magnetic field	Electric field
$v = L \frac{di}{dt}$	$i = C \frac{dv}{dt}$
$i = \frac{1}{L} \int_{t_0}^t v d\tau + i(t_0)$	$v = \frac{1}{C} \int_{t_0}^t i d\tau + v(t_0)$
$i_{DC} \Rightarrow v = 0$	$V_{DC} \Rightarrow i = 0$
Short for DC	Open for DC
$p = Li \frac{di}{dt}$	$p = Cv \frac{dv}{dt}$
$w = \frac{1}{2} Li^2$	$w = \frac{1}{2} Cv^2$

Example

Given $i(t)$, Find $v(t)$



$$i(t) = 0 \quad t \leq 0$$

$$i(t) = 5000t(\text{A}) \quad 0 \leq t \leq 20\mu\text{s}$$

$$i(t) = 0.2 - 5000t(\text{A}) \quad 20\mu\text{s} \leq t \leq 40\mu\text{s}$$

$$i(t) = 0 \quad t \geq 40\mu\text{s}$$

Given

a) $0 \leq t \leq 20\mu\text{s}$

initially
uncharged

$$v = \frac{1}{0.2 \times 10^{-6}} \int_0^t 5000\tau d\tau + 0 \quad \text{General Equation}$$

$$v(t) = 12.5 \times 10^9 t^2 \quad \text{Analytical Expression}$$

$$v(t = 20\mu\text{s}) = 5(\text{V})$$

$t = 20 (\mu\text{s})$ in the above equation

Example

Given $i(t)$, Find $v(t)$ (Contd.)

b) $20\mu s \leq t \leq 40\mu s$ $t_0 = 20\mu s$

$$v = \frac{1}{0.2 \times 10^{-6}} \int_{20\mu s}^t (0.2 - 5000\tau) d\tau + v(t_0) \quad \left. \vphantom{\int} \right\} \text{General Equation}$$

$$v(t_0) = v(20\mu s) = 5(V) \quad \left. \vphantom{v(t_0)} \right\} \text{From part (a)}$$

$$v(t) = -1.25 \times 10^{10} t^2 + 10^6 t - 10 \quad \left. \vphantom{v(t)} \right\} \text{Analytical Expression}$$

$$v(40\mu s) = 10(V)$$

$t = 40 (\mu s)$ in the above equation

Example

Given $i(t)$, Find $v(t)$ (Contd.)

c) $t \geq 40\mu s$ $i(t) = 0$

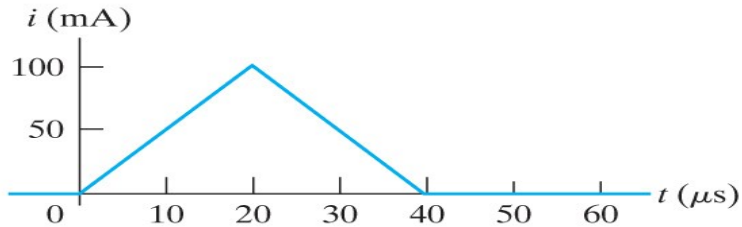
Obtained from part (b)

$$v = \frac{1}{0.2 \times 10^{-6}} \int_{40\mu s}^t (0) d\tau + v(40\mu s) \quad \left. \vphantom{\int} \right\} \text{General Equation}$$

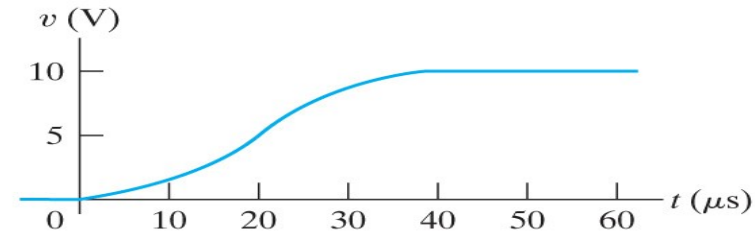
$$v(t) = v(40\mu s) = 10(V)$$

v is constant

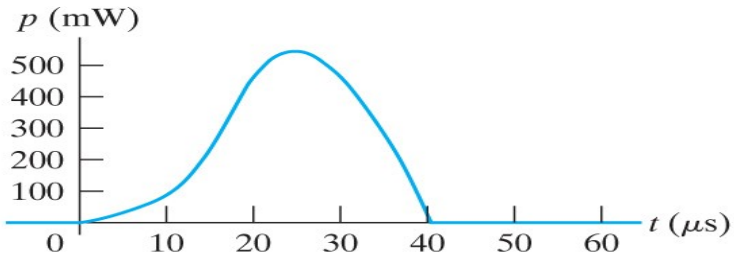
Example Plots



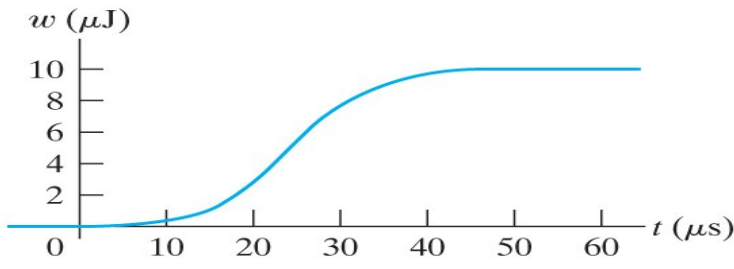
Current through Capacitor



Voltage across Capacitor



Power absorbed by Capacitor



Energy stored in Capacitor

Comments on Example Plots

① Voltage Constant for $t \geq 40(\mu s)$, $i = 0$

② Since v is Constant for $t \geq 40(\mu s)$
 $w = \frac{1}{2}Cv^2$ is Constant for $t \geq 40(\mu s)$

③ $p > 0$ $0 < t < 40(\mu s)$

Power Always Absorbed

*Whole industry
based on this
concept!*

Conclusion: We charged up a Capacitor for 40 μs , and the Capacitor remained charged

Example

Given $i(t)$, Find $v(t)$ (Contd.)

Summary of Results

1	$v(t) = (12.5 \times 10^9)t^2$	$0 \leq t \leq 20(\mu s)$
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2	$v(t) = (-1.25 \times 10^{10})t^2 + (10^6)t - 10$	$20(\mu s) \leq t \leq 40(\mu s)$
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3	$v(t) = v(40\mu s) = 10(V)$	$t \geq 40(\mu s)$
---	-----------------------------	--------------------

Example

Given $i(t)$, Find $v(t)$ (Contd.)

Question: What is the Voltage Across, & the Energy Stored in, the capacitor at $t = 31 \mu s$?

Use expression in (2) on the previous slide

$$\begin{aligned} v(31\mu s) &= (-1.25 \times 10^{10})(31 \times 10^{-6})^2 + (10^6)(31 \times 10^{-6}) - 10 \\ &= -12.0125 + 31 - 10 \end{aligned}$$

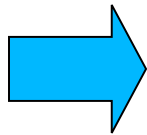
$$v(31\mu s) = 8.9875(V)$$

$$w(31\mu s) = \frac{1}{2} C v^2(31\mu s) = \frac{1}{2} (0.2 \times 10^{-6})(8.9875)^2$$

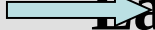
$$w(31\mu s) = 8.078(\mu J)$$

Series and Parallel Combinations

Inductors

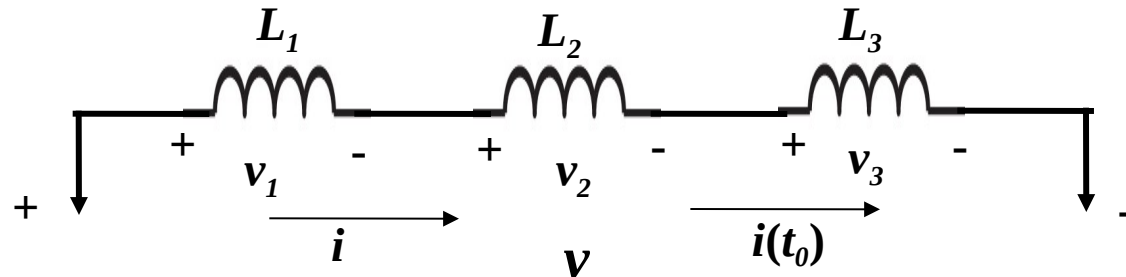


Analogous to Resistors.

Longer Inductor  Larger L values

Due to # of Windings

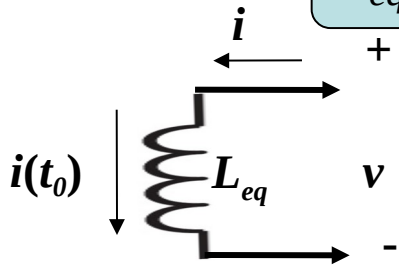
Series



i same through
all inductors

$$L_{eq} = L_1 + L_2 + L_3$$

Add series inductors



$$v = L_{eq} \frac{di}{dt}$$

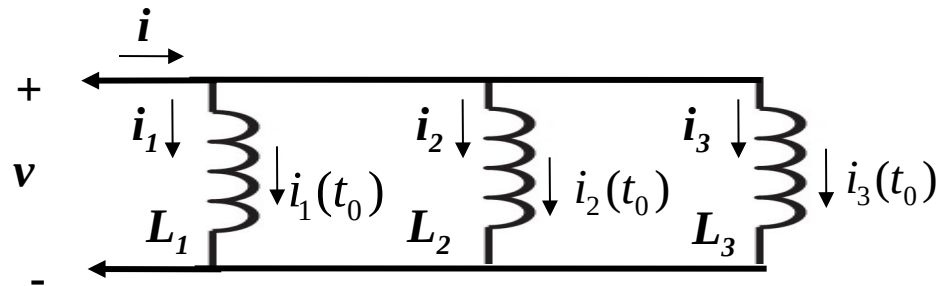
$$v = v_1 + v_2 + v_3$$

Series and Parallel Combinations

(Contd.)

Inductors

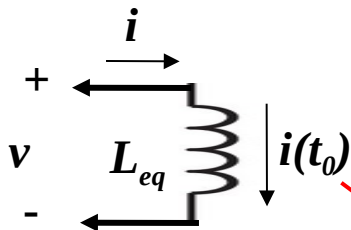
Parallel



v same across all inductors

$$L_{eq} = \left(\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} \right)^{-1}$$

Add reciprocals and invert



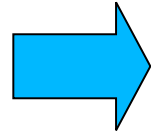
$$i = i_1 + i_2 + i_3 = \frac{1}{L_{eq}} \int_{t_0}^t v d\tau + i(t_0)$$

$$i(t_0) = i_1(t_0) + i_2(t_0) + i_3(t_0)$$

Inductors Combine like Resistors

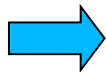
Series and Parallel Combinations

Capacitors



(Contd.)
Dual of Inductor

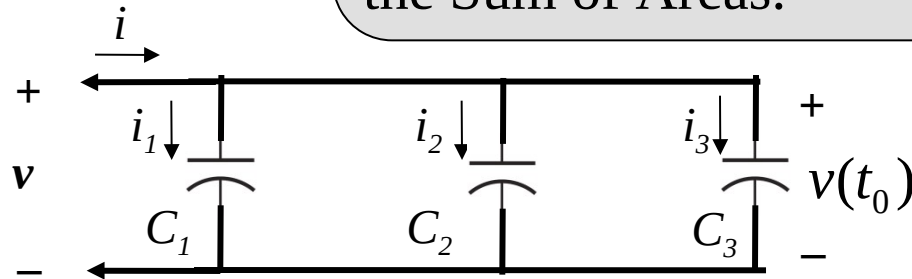
Parallel



Caps Store Charge Between “Plates”.

Larger Area $\rightarrow$ Larger Capacity $\rightarrow$ Larger C

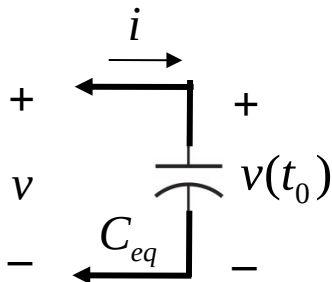
Caps in Parallel Add since Total “Capacity” is Due to the Sum of Areas.



v same across all the Capacitors

$$C_{eq} = C_1 + C_2 + C_3$$

Add in Parallel



$$i = C_{eq} \frac{dv}{dt}$$

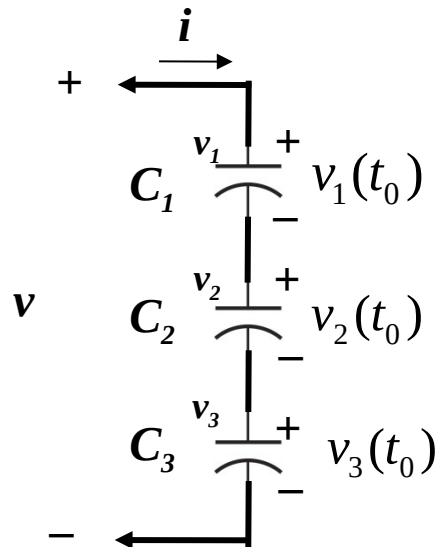
$$i = i_1 + i_2 + i_3$$

Series and Parallel Combinations

(Contd.)

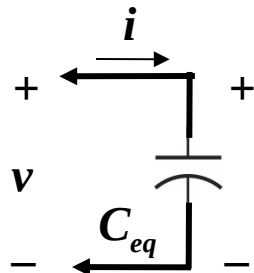
Capacitors

Series



i same through all the Capacitors

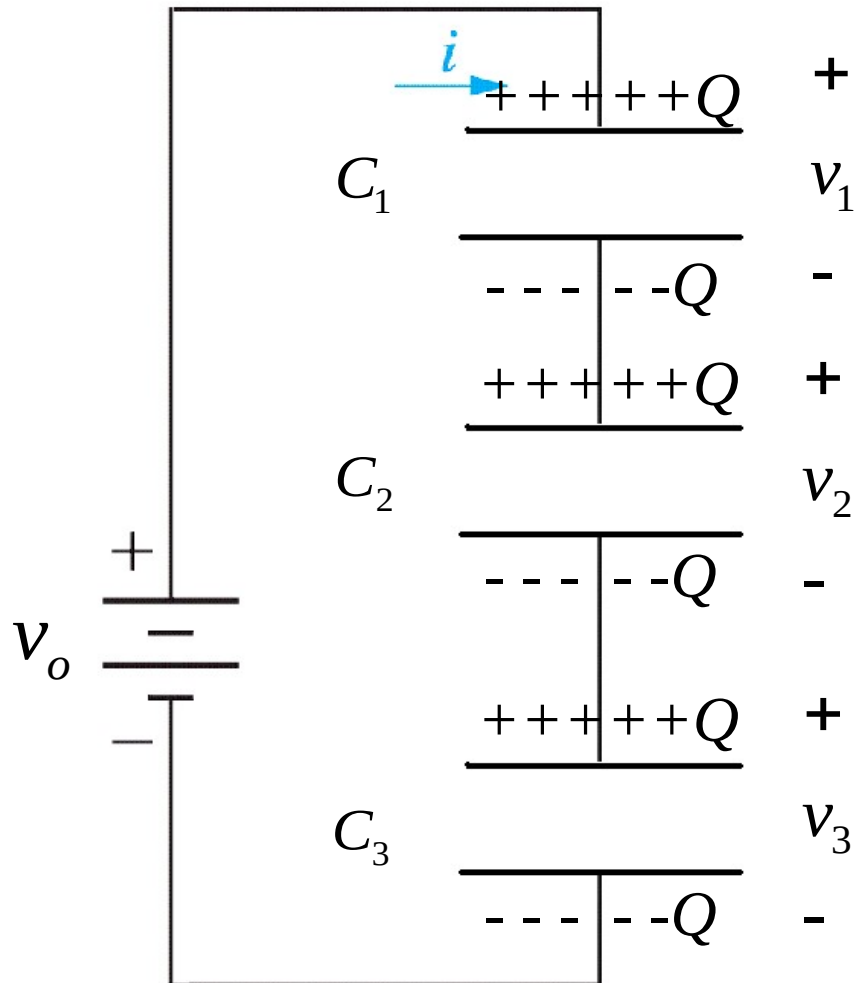
$$C_{eq} = \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)^{-1}$$



$$v = v_1 + v_2 + v_3 = \frac{1}{C_{eq}} \int_{t_0}^t i d\tau + v(t_0)$$

$$v(t_0) = v_1(t_0) + v_2(t_0) + v_3(t_0)$$

“Voltage Division” for Capacitors



Q 's are the same due to charge conservation

$$Q = Cv$$

$$\therefore Q = C_1 v_1 = C_2 v_2 = C_3 v_3$$

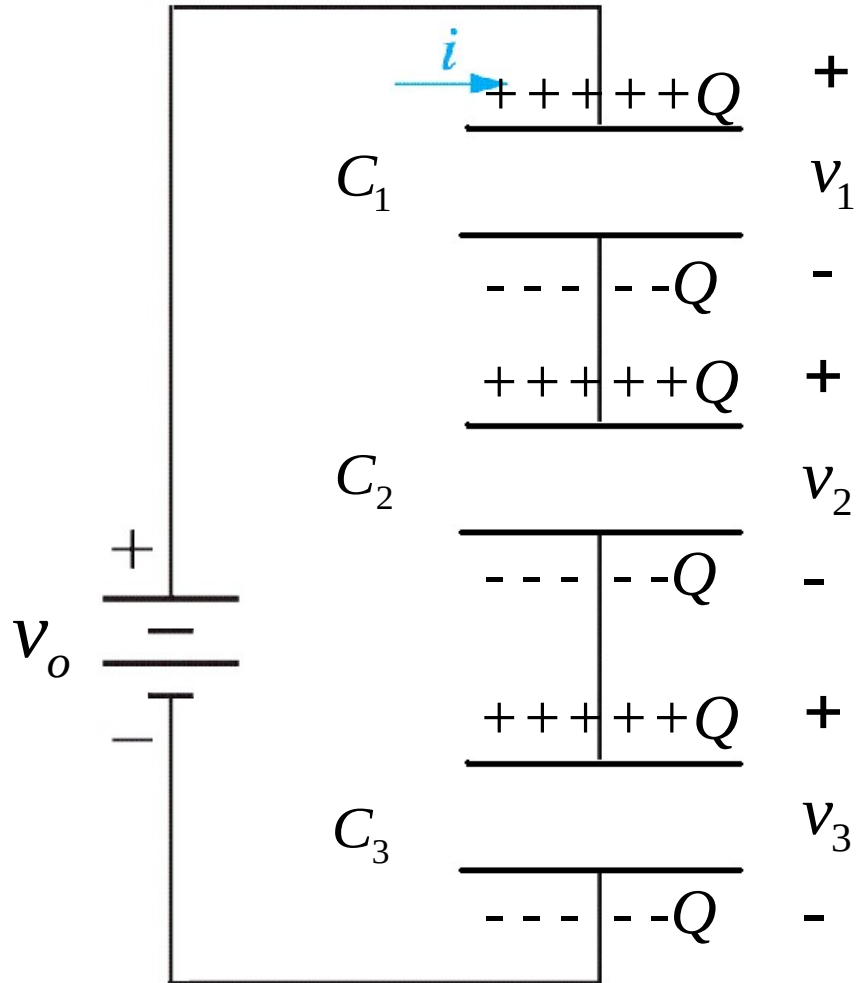
$$Q = C_{eq} v_o$$

$$C_{eq} = \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)^{-1}$$

$$v_o = v_1 + v_2 + v_3$$

“Voltage Division” for Capacitors

(Contd.)



$$Q = C_{eq} v_o \} \text{ From previous slide}$$

$$v_1 = \frac{Q}{C_1} = \frac{C_{eq}}{C_1} v_o$$

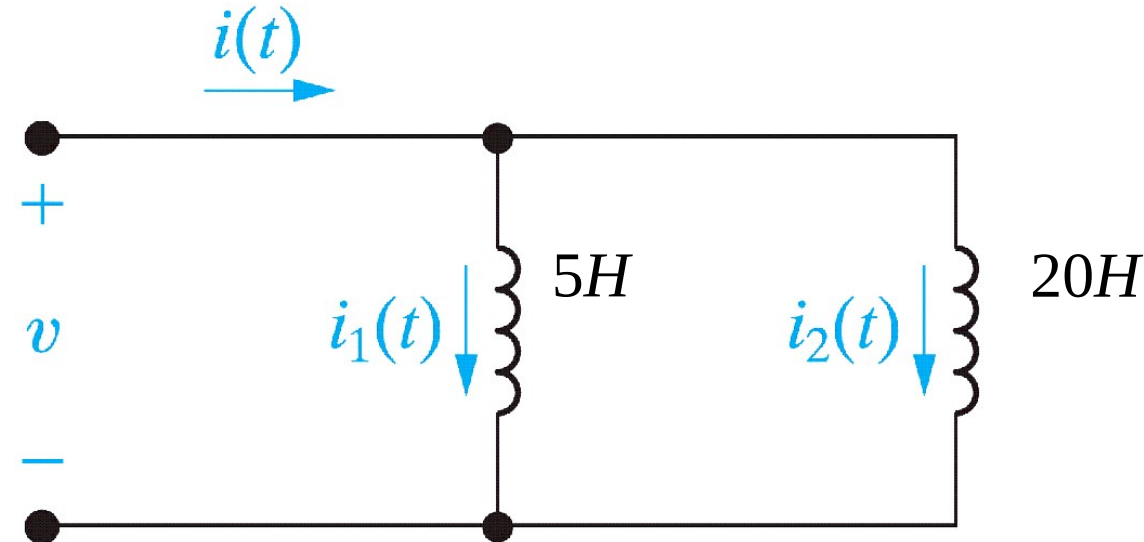
$$v_2 = \frac{Q}{C_2} = \frac{C_{eq}}{C_2} v_o$$

$$v_3 = \frac{Q}{C_3} = \frac{C_{eq}}{C_3} v_o$$

“Voltage Division”

Example

Given $v(t)$, $i_1(0)$, and $i_2(0)$; Find $i(t)$



$$i_1(0) = -2(A)$$

$$i_2(0) = +4(A)$$

$$v = -40e^{-5t}(V) \quad t \geq 0$$

$$\text{a) } L_{eq} = \frac{L_1 L_2}{L_1 + L_2} = \frac{5(20)}{5 + 20} = \frac{100}{25}$$

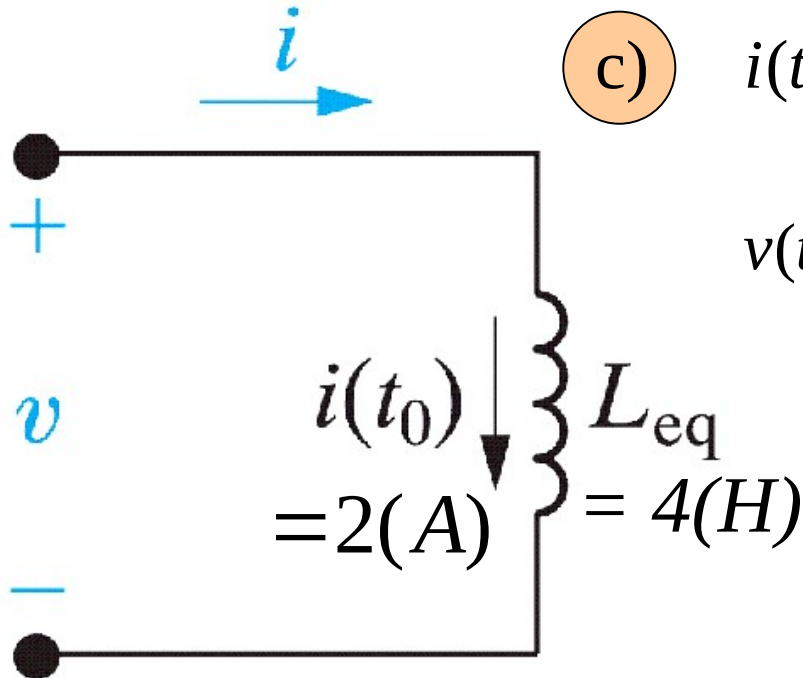
$$L_{eq} = 4(H)$$

$$\text{b) } i(t_o) = i_1(t_o) + i_2(t_o) = -2 + 4$$

$$i(t_o) = +2(A)$$

Example

Given $v(t)$, $i_1(0)$, and $i_2(0)$; Find $i(t)$ (Contd.)



c)
$$i(t) = \frac{1}{L_{eq}} \int_0^t v(\tau) d\tau + i(t_0) \quad \left. \vphantom{\int_0^t} \right\} \text{General Equation}$$

$$v(t) = -40e^{-5t} (V)$$

Plug in $v(t)$ and $i(t_0)$

$$i(t) = \frac{1}{4} \int_0^t (-40e^{-5\tau}) d\tau + 2$$

$$i(t) = -10 \left[-\frac{1}{5} e^{-5\tau} \right]_0^t + 2 = 2 \left[e^{-5t} - e^0 \right] + 2 \quad \left. \vphantom{\int_0^t} \right\} \text{Integrate and Simplify}$$

$$i(t) = 2e^{-5t} \quad t \geq 0$$